Answers and talk notes

Ratio Reasoning: Convert Measurements within the Same System

What your student is learning

Tonight we convert metric units — meters, grams, liters — using powers of ten, and check the answer makes sense.

Answers

Ask your student to explain before you look.

1. **C.** 1,000. The prefix kilo means 1,000, so 1 kilometer = 1,000 meters.
2. **A.** 5,000. 5 × 1,000 = 5,000, so 5 kilometers = 5,000 meters.
3. **B.** 3. 3,000 ÷ 1,000 = 3, so 3,000 grams = 3 kilograms.
4. **D.** 400. 4 × 100 = 400, so 4 meters = 400 centimeters.
5. **A.** 2,000. 2 × 1,000 = 2,000, so 2 liters = 2,000 milliliters.
6. **C.** 0.65. 650 ÷ 1,000 = 0.65, so 650 millimeters = 0.65 meters.

Talk about it

Are you changing to a smaller unit or a bigger unit, and what does that tell you to do?

Home mission

Find a bottle labeled 1 L or 1,000 mL. Say: "The cooler holds 6 liters." Ask your student how many bottles like this fill it. Then ask how many milliliters that is in all.

When they are stuck, ask

- Which unit is smaller — the one we have, or the one we want?
- Will there be more of them or fewer of them?
- Does the answer look reasonable for that unit?

Try not to say

- Just multiply, that is how conversions work.
- Move the decimal two places, it is usually two.
- Do not worry about whether the answer makes sense.